

WASP-52b

R-1669

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_200811 · created 2026-08-02T10:21:50+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459072.7628 +/- 0.0051 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.213 +/- 0.042
Transit depth (Rp/Rs)^2	4.54 +/- 1.78 [%]
Orbital Inclination (inc)	82.8 +/- 2.29
Airmass coefficient 1 (a1)	0.949 +/- 0.026
Airmass coefficient 2 (a2)	0.0383 +/- 0.0072
Scatter in the residuals of the lightcurve fit is	2.78 %
Best Comparison Star	#1 - [144, 330]
Optimal Aperture	3.23
Optimal Annulus	11.13
Transit Duration (day)	0.053 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.213 ± 0.042 vs 0.1646 (deviation 1.15σ)
Duration fit vs published	0.053 d vs 0.0758 d (deviation 0.78σ)
Mid-time O–C	-12.6 min
Depth SNR	5.1
Residual scatter	2.78 %

Light curve

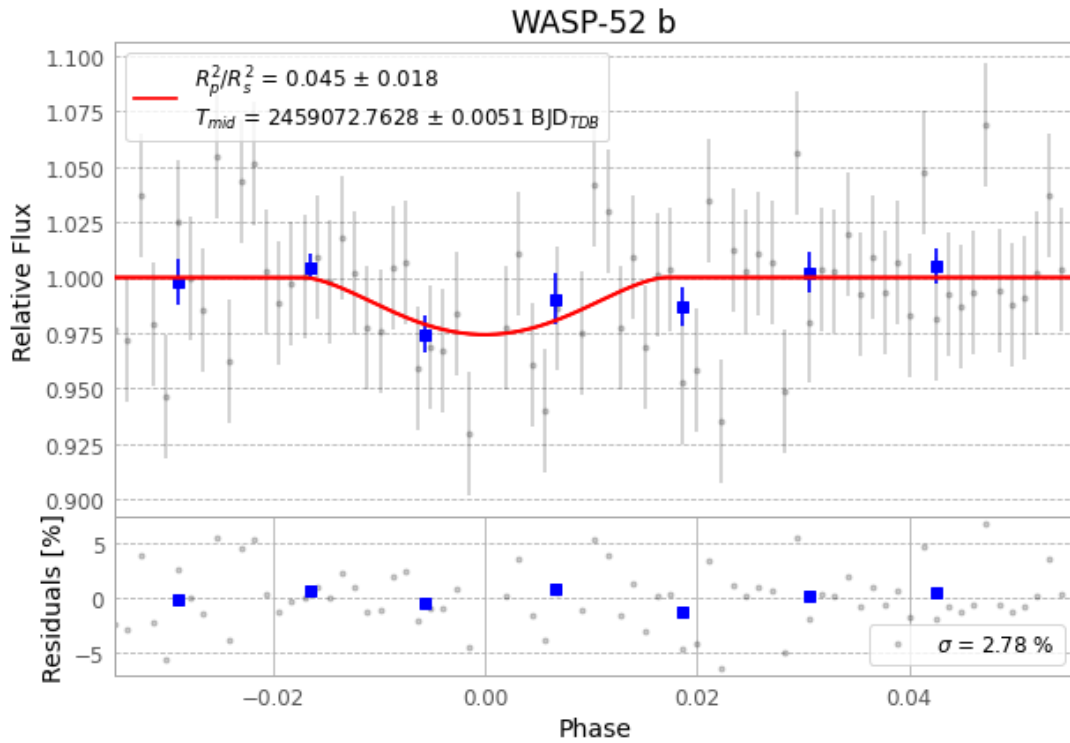


Figure 1 — FinalLightCurve_WASP-52 b_2020-08-10.png (SHA-256 cee52f639845506c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [174, 146], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2fee957497817d66...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-52200811044212.FITS → WASP-52200811083012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-200811.log (f6f36c774f7b...), FinalLightCurve_WASP-52 b_2020-08-10.png (cee52f639845...), FinalLightCurve_WASP-52 b_2020-08-10.csv (9487e6b6a8a7...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 10:21Z.